

Are Turkish neighborhoods becoming politically more uniform?

Five parliamentary elections' worth of ballot boxes, about 919,000 of them, measured for political sorting inside Turkey's mahalles, 2011-2023.

1. The question

In 2020, Özer and Taşkın published a study of how one gated community in Tuzla voted ("Ayıran Duvarlar, Dönüşen Sandıklar," MSGSÜ Sosyal Bilimler Dergisi). Official results could not answer their question directly: the finest published unit is the sandık, the individual ballot box, and nothing in the data says which side of a wall a box sits on. So they went to the field and interviewed local party organizations to learn which sandık numbers belonged to the complex. The result confirmed their suspicion: boxes a wall apart voted sharply differently. One neighborhood on paper; two political worlds on the ground.

That study covered one complex, by hand. This paper computes the same quantity for the whole country, across twelve years: are like-minded voters increasingly concentrated together? Are neighborhoods becoming internally more homogeneous, or more divided? This is the Turkish version of the American "Big Sort" question, and the nationwide, longitudinal answer did not exist before this analysis.

The short version:

Within neighborhoods that stayed comparable across all five elections from 2011 to 2023, the vote shares of the three long-established parties (AKP, CHP, and MHP) became measurably less dispersed across the ballot boxes of the same neighborhood. Turkish mahalles became modestly more internally uniform, with the clearest decline for CHP.

The long version is a demonstration of what it takes to make that sentence survive: three artifact traps identified and controlled, and one seductive false finding caught and killed. That methodological story is a result in its own right.

2. The data

The raw material is the box-level results of the five parliamentary (milletvekili) general elections: June 2011, June 2015, November 2015, June 2018, and May 2023. After cleaning, each election contributes between roughly 174,000 and 199,000 usable ballot boxes, about 919,000 box-level records in all, covering electorates growing from 46 million registered voters in 2011 to 56 million in 2023.

Each box record carries its geography: province (il), district (ilçe), neighborhood (mahalle), and box number. The mahalle is the key. A mahalle typically contains a handful to a few dozen boxes, each serving a few hundred registered voters assigned by address. We never see an individual's vote, but we do see, at address-block resolution, how uniformly or unevenly a party's support is spread inside each neighborhood. That within-mahalle granularity is what mahalle-aggregated studies cannot compute. It is the distinctive asset here, and no other nationwide Turkish analysis has used it this way.

Two limits of the data, stated up front:

- For 2011, 2018, and 2023, the mahalle is the finest geographic key. Only the two 2015 elections carry school/building names. Boxes within a mahalle can be compared to each other, but not placed on a map more finely than their mahalle.
- Everything here is areal. We measure boxes and neighborhoods, not people. Nothing in this paper says any voter changed their mind, their party, or their address.

Cleaning this data is its own small war, and two details deserve mention because they silently corrupt results if ignored. First, Turkish text: the dotted and dotless *ı*/*ı* do not survive standard lowercasing. A naive `lower()` turns *İSTANBUL* and *ISPARTA* into keys that no longer match their own records, so every place-name join runs through a Turkish-aware normalizer, enforced by tests. (The same bug bit this project's own chart labels during figure production, which says something about how easily it is reintroduced.) Second, vote-share denominators: shares are computed against reported valid votes (*geçerli oy*), with an explicit residual column for small parties and independents, rather than against the sum of major-

party columns. The latter silently inflates shares in exactly the spatially correlated way that would fake a sorting trend. The 2018 and 2023 alliance-era files keep alliance lump votes in separate, clearly labeled columns, never folded into party totals.

3. The measure

For each mahalle, each election, and each party, we ask: how much do this party's vote shares vary across the ballot boxes of this one neighborhood?

Concretely: take each box's party share (party votes $\div$ valid votes) and compute the spread across the mahalle's boxes, a standard deviation weighted by box size. A mahalle where AKP polls 40% at every box has spread zero: a politically uniform neighborhood, whatever its lean. A mahalle where AKP polls 20% at half the boxes and 60% at the other half is internally divided. That is the gated-community pattern, visible without knowing where the wall is.

Raw spread has a flaw for trend purposes: it scales with the party's level of support. A party sliding from 45% to 10% will mechanically show smaller absolute spread even if its geography is unchanged. So the workhorse statistic is the coefficient of variation (CV): the spread divided by the party's mean share in that mahalle. CV asks how dispersed a party's support is relative to its own size, which is the question we actually mean. National summaries weight mahalles by registered voters, so the series describes where voters actually live rather than treating a 300-voter village and a 40,000-voter urban mahalle as equal observations.

4. Three ways to fool yourself, and the ladder built against them

The naive computation (CV per mahalle, national average per election, draw the line) produces a satisfying decline. We treated that line as a suspect, not a finding, and tested three artifact channels. It survived all three. Most quick analyses of this data would not have run the tests.

Trap 1: the crowded field. The party system fragmented between 2011 and 2023; more parties means lower shares, which distorts raw spreads. Using CV rather than raw SD addresses the mechanical part, and the decline survives: comparing only the parties present throughout (AKP, CHP, MHP), the voter-weighted national CV falls from 2011 to 2023 for all three (AKP 0.165→0.134, CHP 0.325→0.177, MHP 0.351→0.284, uncontrolled series).

Trap 2: box counts. The number of boxes per mahalle changed over time, and measured dispersion correlates with box count (more boxes, more room to disagree, partly a small-sample effect). This correlation was material in 11 party-election cells, so box-count stability became an explicit control condition rather than a footnote.

Trap 3: boundary churn, the big one. Mahalle identifiers do not simply carry over between elections. Looking up each mahalle of one election by name in the next: from 2011 to June 2015, only 42% of mahalle names can be re-found (the 2012–2014 metropolitan-municipality reform re-labeled huge numbers of villages as mahalles; the data records the churn, not its cause). From 2015 onward, names mostly match (~85–87%), but the matched names hold only about half the registered voters, meaning the unmatched names are on average the much more populous places. A "trend" computed on whatever matches year to year would silently swap large parts of the electorate in and out and call the result change.

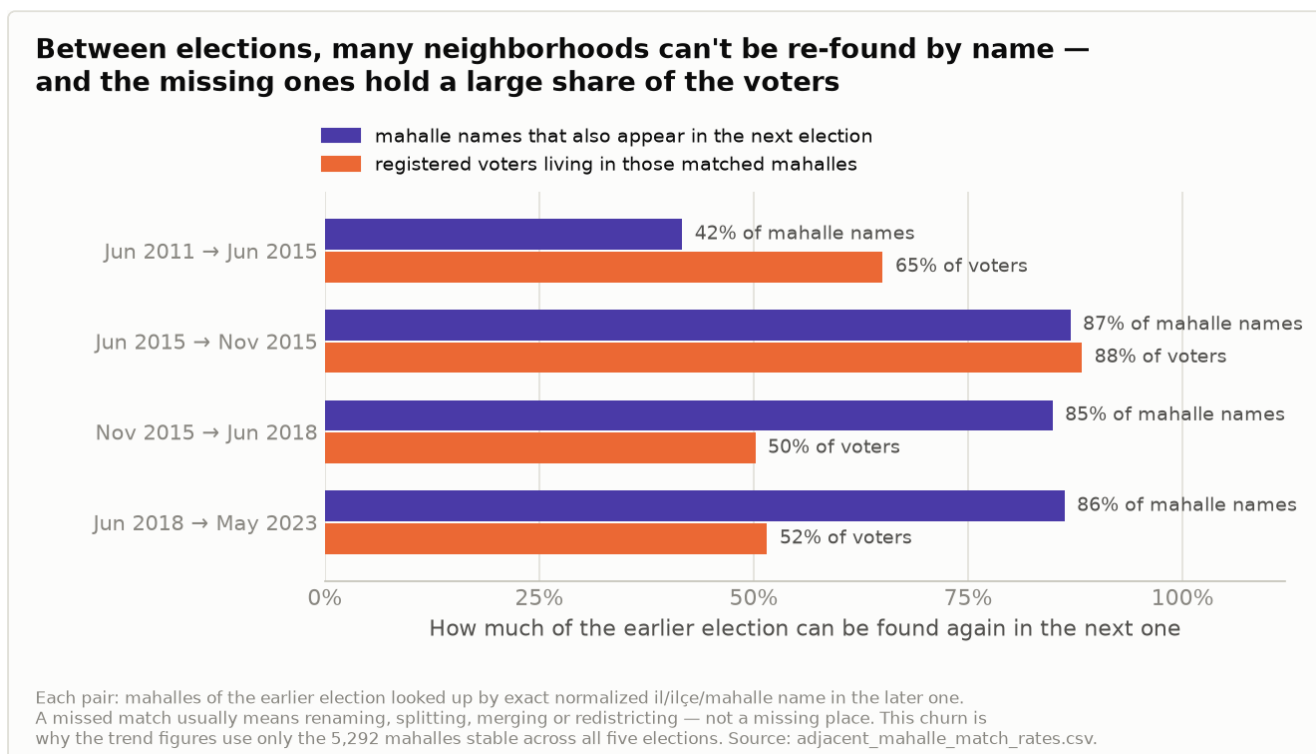


Figure 5. Adjacent-election continuity: share of mahalle names re-found in the next election (violet) and of registered voters living in the matched mahalles (orange).

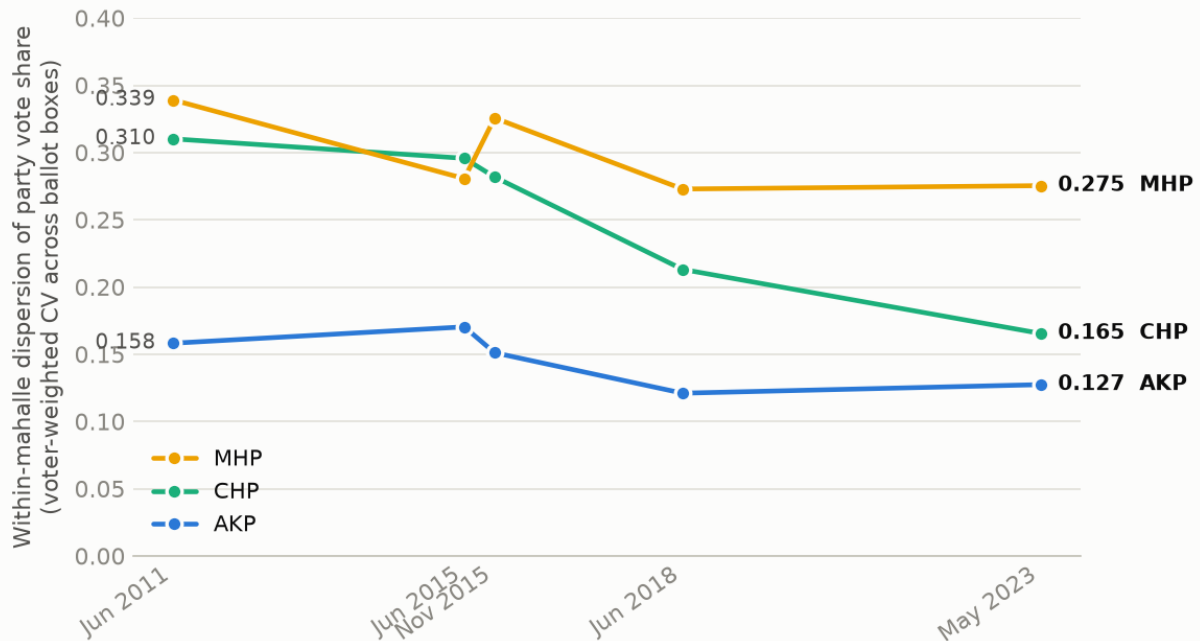
The response is a ladder of increasingly strict conditions, ending in **condition 4**: only mahalles whose province/district/mahalle key exists in all five elections, and whose box count stays stable across the whole run (at least 2 boxes every time; max/min ratio ≤ 1.5 , or an absolute range of at most 2). That leaves **5,292 full-chain stable mahalles**, covering about 9.5 million registered voters in 2011 and 11.3 million in 2023, roughly a fifth of the electorate, deliberately traded for comparability. A province-anchored version of the statistic serves as an independent sanity check on direction.

5. The finding

On the strict condition-4 subset, voter-weighted within-mahalle CV:

PARTY	2011	2023	DIRECTION
AKP	0.158	0.127	decline
CHP	0.310	0.165	strong decline
MHP	0.339	0.275	decline

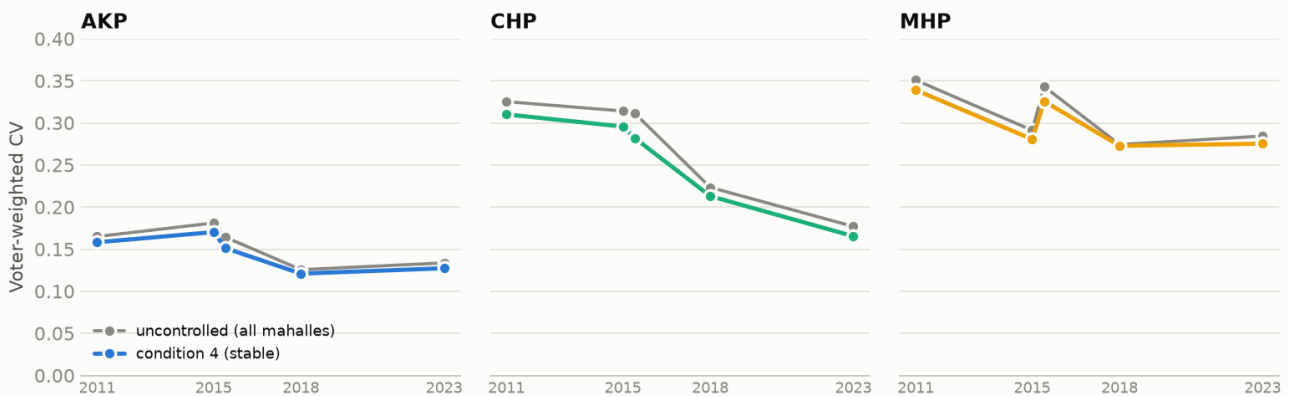
Within stable mahalles, AKP, CHP and MHP vote shares became less dispersed across ballot boxes, 2011-2023



Strict control (condition 4): 5,292 full-chain stable mahalles with stable box counts, 9.5M (2011) to 11.3M (2023) registered voters. Areal measure over ballot boxes; says nothing about individual voters. Source: m3_four_condition_trends.csv, voter-weighted.

Figure 1. Within-mahalle dispersion (voter-weighted CV) for AKP, CHP, MHP on the 5,292 full-chain stable mahalles of condition 4.

The decline is not a boundary-churn artifact: it survives the strict stable-geography control



Gray: all mahalles, no controls. Color: condition 4 — full-chain stable mahalle keys and stable box counts. Source: m3_four_condition_trends.csv.

Figure 3. Uncontrolled national series (gray) versus condition-4 stable-geography series (color). The decline survives the strictest control.

Three things about this result.

It survives the controls, visibly but not trivially. The uncontrolled and condition-4 series differ (the control is doing real work) but the direction holds in both, for all three parties. The decline is not an artifact of boundary churn or shifting box counts. That is a statement most snapshot analyses of Turkish electoral geography cannot make, because they do not construct the panel that makes the test possible.

The metric is not a one-way ratchet. MHP's raw within-mahalle spread rises from 2011 to June 2015 (0.033→0.034) before falling in later elections. A pure mechanical-compression artifact would not let a party move against the tide; that it can is direct evidence the measure captures real, party-specific geography.

The magnitude is modest, and we say so precisely. CHP's within-mahalle CV roughly halved over twelve years; AKP's and MHP's declined by about a fifth. This is a real, consistent compression of within-neighborhood political variation among the established parties, not a transformation of the country. "Turkey homogenized" is not the claim, and the data as analyzed so far would not support it.

6. The finding that wasn't: how the HDP/YSP artifact was caught

For a while, the analysis appeared to contain a second, more dramatic story. The Kurdish-movement lineage (HDP in 2015–2018, running as Yeşil Sol/YSP in 2023, a well-documented continuity handled explicitly by the party-code crosswalk) showed the opposite trend: within-mahalle CV rising from 0.524 in June 2015 to 0.630 in 2023 on the same strict stable subset. Rising internal fragmentation for one party, against homogenization for the big three, in a politically charged corner of the data. An irresistible headline.

It is wrong, and the way it is wrong is instructive. A coefficient of variation explodes where a party's support approaches zero: with a 1% mean share, trivial absolute differences between boxes become enormous relative spread. If the apparent rise were real neighborhood-level fragmentation, it would survive restricting to mahalles where the party actually has support. It does not. It reverses:

MAHALLES INCLUDED (CONDITION-4 SUBSET)	JUN 2015 CV	2023 CV	DIRECTION
All	0.524	0.630	rises
Party $\geq$ 5% support	0.341	0.243	declines
Party $\geq$ 10% support	0.268	0.168	declines
Party $\geq$ 20% support	0.157	0.101	declines

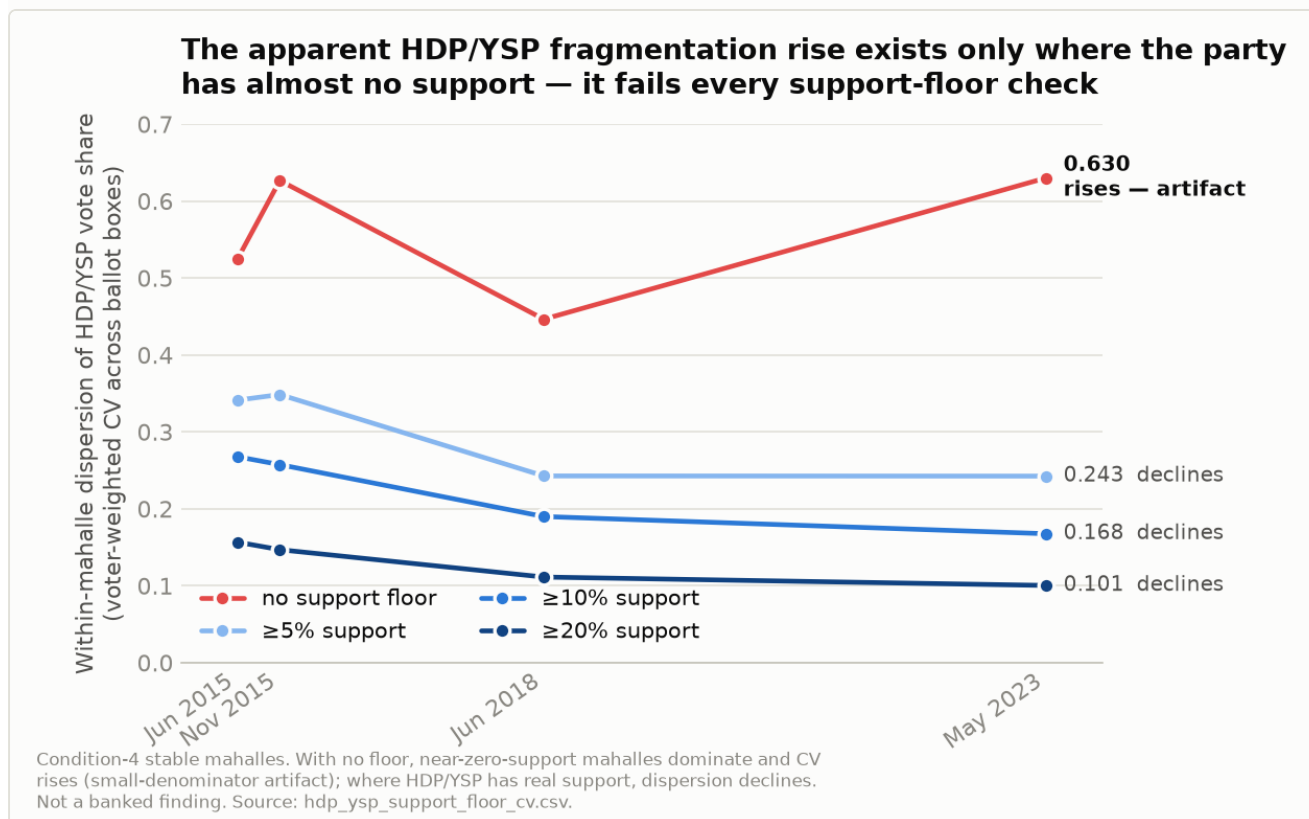


Figure 2. HDP/YSP within-mahalle CV on the condition-4 subset, with and without support floors. The rise exists only with no floor — a small-denominator artifact, not a finding.

Wherever HDP/YSP has a real presence, its neighborhoods homogenized like everyone else's. The nationwide "rise" lives entirely in near-zero-support mahalles: a small-denominator artifact, not sorting. Two corroborating checks: the independent province-anchored series never showed the rise (roughly flat-to-down while the strict mahalle series rose, which is the divergence that triggered scrutiny), and turnout-composition checks in the southeast found no strong turnout-driven explanation to rescue a substantive reading.

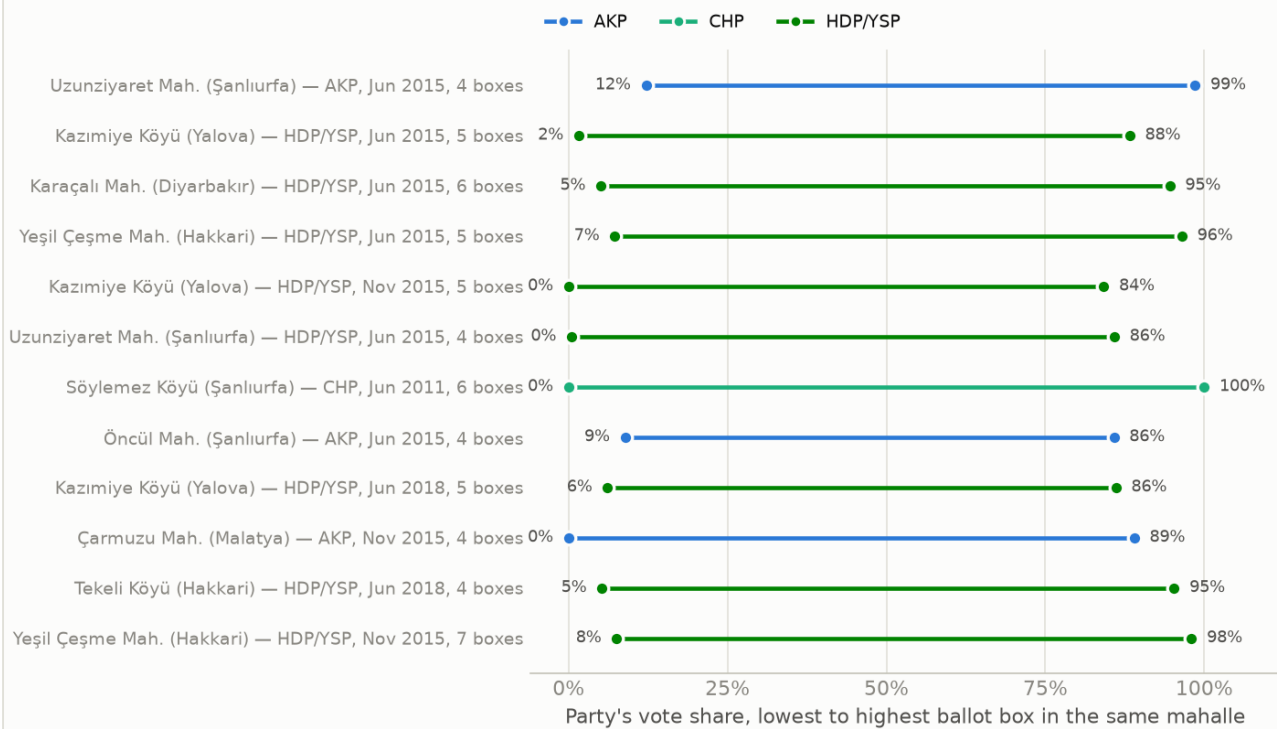
The fragmentation claim is dead, and this paper does not make it. It was the most tempting result in the project, which is exactly why it received the most hostile examination. Catching it is one of the paper's contributions: any analysis of a regionally concentrated party at fine resolution will hit this trap, and most will not notice.

7. Neighborhoods split down the middle

The national trend is one use of the data. The other is the automated version of the gated-community study: rank every party-mahalle-election cell by internal spread and look at the extremes. What Özer and Taşkın established for one complex through fieldwork, this method surfaces nationwide from public data alone.

To keep tiny-box noise out, the flagship list requires at least 4 boxes and at least 150 registered voters per box; the filtered top-100 overlaps the unfiltered one by 81%, so the extremes are not merely small-denominator noise. The cases are striking. In Uzunziyaret Mahallesi (Siverek, Şanlıurfa) in June 2015, AKP polled 12% at one ballot box and 99% at another: four boxes, one neighborhood, 885 registered voters. In Kazımiye Köyü (Yalova), HDP ranged from 2% to 88% across five boxes, and the same place reappears in the extreme list in November 2015 and 2018. Boxes are filled by address, so splits like these are shadows of real social boundaries (a street, a settlement pattern, perhaps a wall) visible in public data without anyone conducting an interview.

Extreme within-mahalle splits: the same party and neighborhood, very different boxes



Top 12 party-mahalle-election cells by within-mahalle share SD, after noise filters (≥ 4 boxes, ≥ 150 registered voters per box). Local micro-fragmentation examples, not national-trend evidence. Source: top_fragmented_filtered.csv.

Figure 4. The most internally split party-mahalle-election cells after noise filters (≥ 4 boxes, ≥ 150 registered voters per box): lowest and highest box share within one neighborhood.

These are local portraits, not trend evidence: a list of extremes is by construction unrepresentative. Their role is to make concrete what within-mahalle dispersion means, and to flag candidate sites for local inspection.

8. The map, and the spatial question it answers

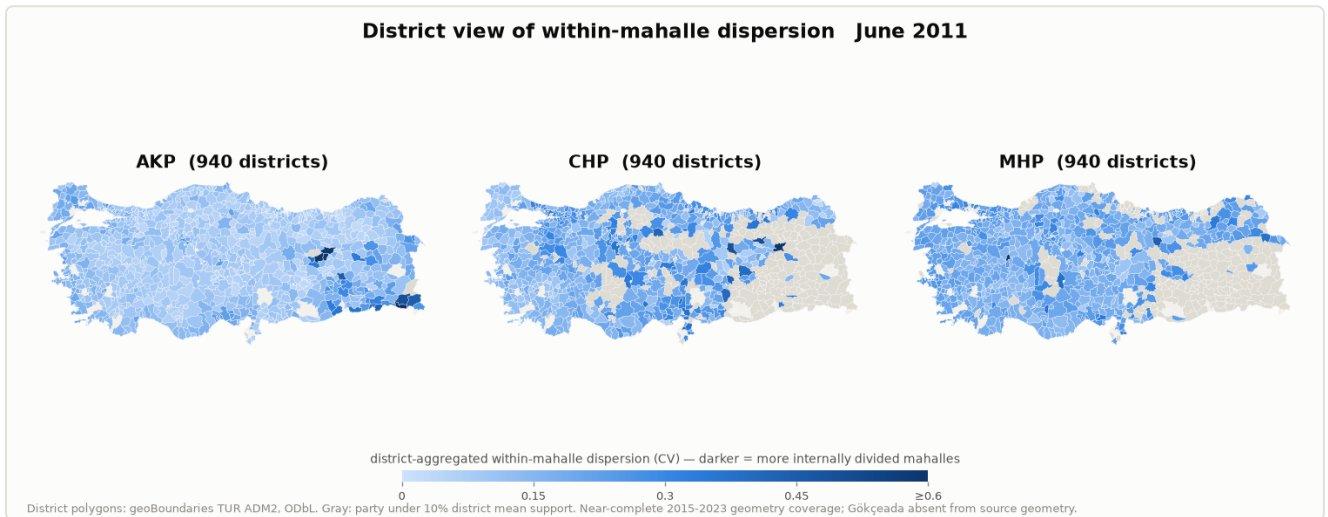


Figure 6a. District aggregation of within-mahalle dispersion per party, June 2011 (first frame of the animation). Gray = under 10% district mean support or no party list.

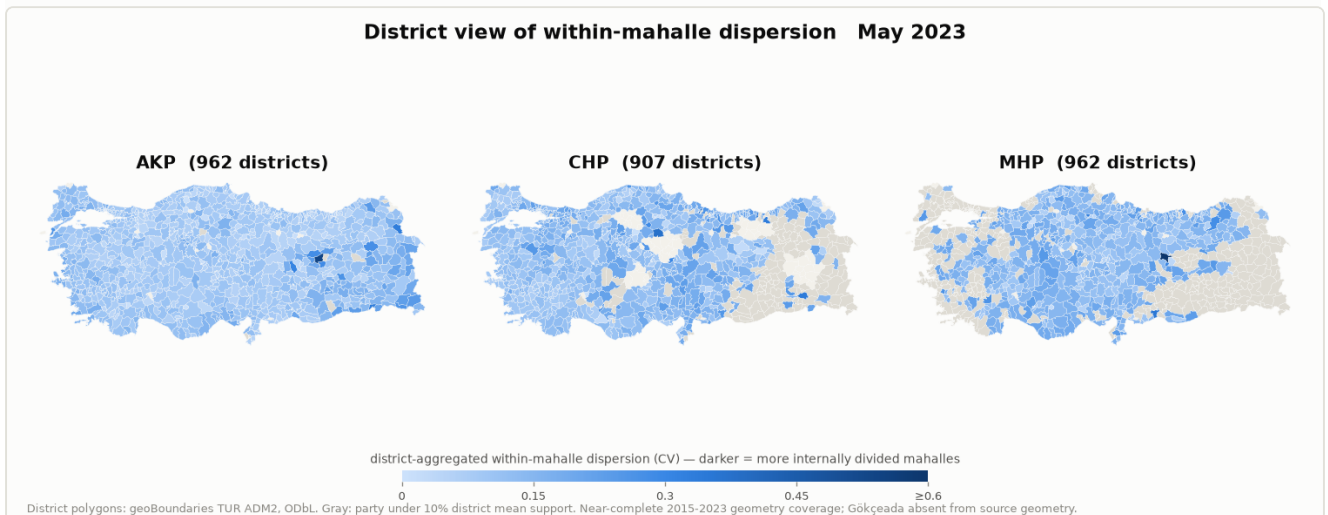


Figure 6b. The same district maps, May 2023 (last frame). The animated version steps through all five elections.

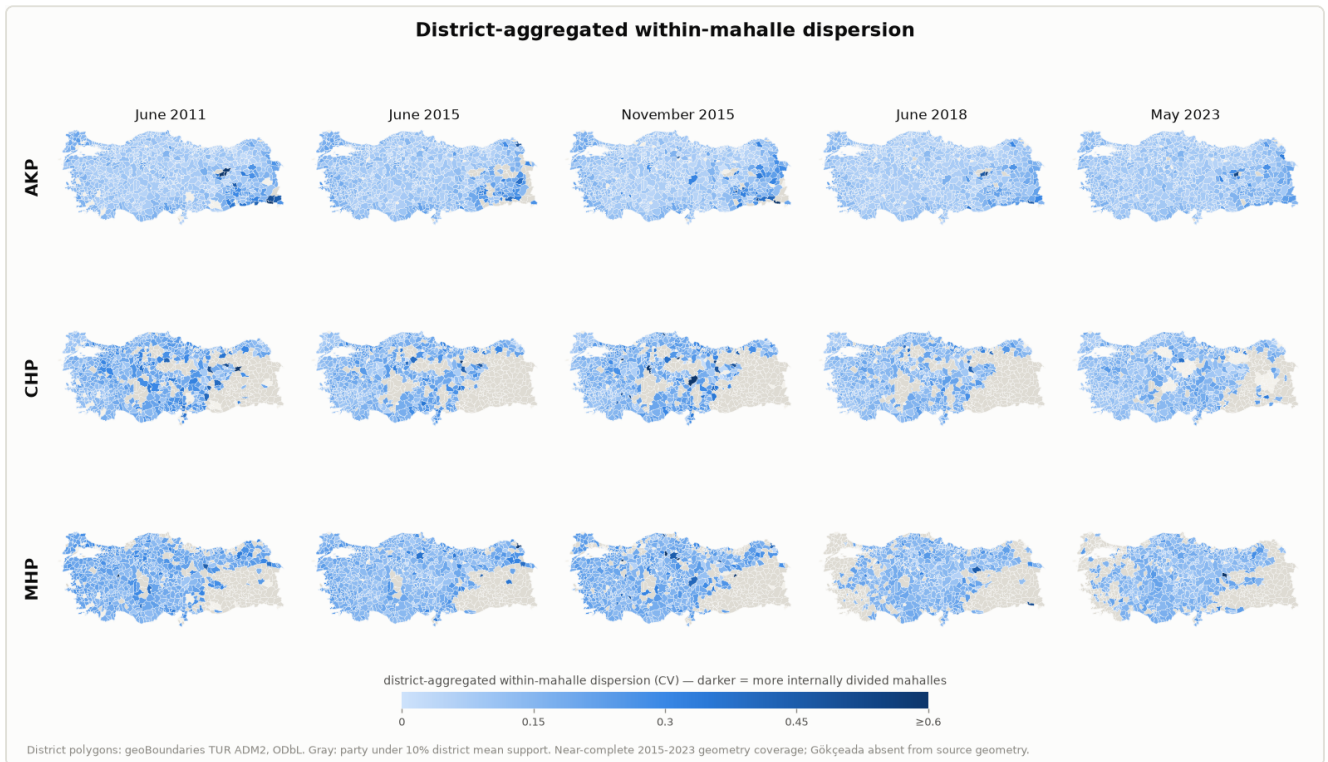


Figure 6c. All five elections as a static district grid — rows fade toward lighter (more internally uniform mahalles within districts).

The map everyone wants, every mahalle shaded by its internal division and animated over five elections, requires national mahalle boundary polygons, which do not exist in this project (see §9). The most granular defensible filled map is district-level: each ilçe polygon shaded by the voter-weighted within-mahalle dispersion of the mahalles inside it, per party, per election. District geometry comes from geoBoundaries' Turkey ADM2 polygons; for 2015–2023 the join is effectively complete (only Gökçeada is missing from the source geometry), while 2011 is shown with a stronger historical-boundary caveat.

At this resolution the finding is visible in a more local way than a province map: the CHP panels lighten distinctly from 2011 to 2023, AKP remains mostly low with pockets of higher within-mahalle dispersion, and MHP fades more modestly. Districts are masked gray where a party polls under 10% (the same small-denominator problem as §6: CV can look dark when the party's mean share is small, and painting those cases would reintroduce the exact artifact we killed) and where a party has no usable list or geometry for that panel. This support-floor caveat is part of the figure's interpretation, not just a web-map interface detail.

The map is a descriptive companion to the trend result, computed as district aggregations of mahalle-level dispersion; it illustrates the finding rather than independently proving it.

We also answered the natural next spatial question rather than leaving it open: if boxes inside stable mahalles became more uniform, did stable mahalles also form stronger geographic clusters with nearby mahalles? Using OSM place-node centroids for the condition-4 stable subset, longitudinal FDR-screened LISA transition tests show no robust 2011-to-2023 change in spatial clustering for AKP, CHP, or MHP. AKP shows endpoint transition structure under one tested weight rule ($k=8$) but not under the primary, $k=6$, or 10 km-cap rules; CHP and MHP show no robust start-end transition signals. The coverage check makes this a clean exploratory null rather than a data-sparsity artifact: the centroid join covers 4,253 of 5,292 stable mahalle keys, 85.3% of 2023 registered voters in the stable subset, and about 90.7% of southeast voters in that subset.

This null sharpens the main result rather than weakening it. Turkish electoral geography has strong, stable spatial structure; Gençkaya and Bakırcı (2022), clustering the 2018 election at district level, found the same three-cluster pattern (AKP/MHP heartland, CHP coast, HDP southeast) that province-level studies had reported for two decades. What that snapshot literature cannot ask, and this project can, is whether the structure is moving. The answer is: the arrangement of the map did not measurably re-sort between 2011 and 2023, while neighborhoods consolidated internally. Consolidation without re-clustering is a more precise account of the decade than either claim alone. HDP/YSP remains more fragile than the other parties in this spatial layer, because its fine-resolution results depend on support floors and spatial-weight choices.

9. What this establishes, and its boundaries

The caveats are part of the result, not disclaimers around it.

- **Areal, not individual.** This is sorting of ballot boxes within neighborhoods. Whether it reflects residential moves, registration changes, local conversion, generational turnover, or something else is not identified. No claim about individual behavior is made or implied.

- **Modest magnitude.** Falling within-mahalle CV for three established parties, strongest for CHP. Not "Turkey transformed."
- **The stable subset is not a random sample.** Condition 4's 5,292 mahalles are places whose names and box counts stayed put. The continuity diagnostics show directly that the excluded mahalles are on average the more populous, metropolitan places: the matched names between elections hold only about half the registered voters. The trade of coverage for comparability is explicit, and the uncontrolled series (which agrees in direction, on ~19,000–24,000 mahalles per election) is always shown alongside.
- **Stable controls match mahalles, not boxes.** Within a stable mahalle, individual boxes are still renumbered and reshuffled between elections; the panel is at neighborhood grain, not box grain.
- **Resolution limits are unresolved (MAUP).** All dispersion is measured within administrative mahalle units. The 2015 elections' school names would allow a finer-resolution recomputation to test how much the unit choice matters; that calibration has not been done.
- **This extends prior work.** Spatial autocorrelation studies of Turkish voting exist at province and district level, almost entirely single-election snapshots, including the district-level K-means clustering of Gençkaya and Bakırcı (2022); one mahalle-level GIS study covers a single province (Hatay, 2018); district-divergence work builds on the Johnston et al. spatial-polarization tradition; and Özer and Taşkın's gated-community study supplied the box-level idea. What none of them do, and this project does, is measure the question nationwide, longitudinally, at box-within-mahalle grain, on a matched stable-geography panel with explicit artifact controls. The contribution is turning a scattered set of snapshots and case studies into a measured national trend.

10. Where this could go next

Three natural extensions, none required for the result above: geocoding the 2015 school names to calibrate how much the mahalle aggregation costs and to build a nationwide wall detector for that year; adding richer local context for the

extreme split-mahalle cases; and using registration and turnout dynamics to probe mechanism, i.e. whether homogenization travels with population churn or happens among stable populations. Each is a harder project than what is banked here, and the panel built for this paper is the foundation any of them would start from.

Appendix: provenance

Every number in this paper traces to a file in the analysis repository, and every figure to a script:

- Cleaned data: `outputs/cleaned_mv/` (five elections; conformance reports alongside). Cleaning contract and tests: `CLEANING_SPEC.md`, `tests/test_cleaning_contract.py` (passing, 4/4, verified this session).
- Condition-4 trend and coverage (§5, Figs 1 & 3): `outputs/m3_boundary_boxcount_control/m3_four_condition_trends.csv`, `m3_condition_coverage.csv`; control definitions in `m3_audit.json`.
- Uncontrolled CV check and box-count correlations (§4): `outputs/pre_m3_artifact_checks/artifact_check_verdict.json`.
- HDP/YSP resolution (§6, Fig 2): `outputs/hdp_ysp_resolution/hdp_ysp_support_floor_cv.csv`, `hdp_ysp_verdict.json`.
- Fragmented-mahalle list (§7, Fig 4): `outputs/pre_m3_artifact_checks/top_fragmented_filtered.csv`.
- Continuity diagnostics (§4, Fig 5): `outputs/cleaning_audits/adjacent_mahalle_match_rates.csv`.
- Map (§8, Fig 6): `outputs/geo_join/district_map_values.csv`, geometry from local `geoBoundaries-TUR-ADM2-all.zip`, join audit `outputs/geo_join/geoboundaries_adm2_join_coverage.csv`, rendered by `scripts/make_district_choropleth_map.py`.
- M2 longitudinal LISA null (§8): `docs/m2_lisa_transitions_status.md`, `outputs/m2_lisa_transitions/`, `outputs/m2_closeout_checks/`, `scripts/m2_lisa_transitions.py`, and `scripts/m2_closeout_checks.py`.

- Figures: `scripts/make_findings_figures.py` , `scripts/make_district_choropleth_map.py` , and `scripts/make_homogenization_map.py` → `outputs/figures/` .

Findings register: `FINDINGS_BANK.md` . No claim in this paper is stronger than its entry there.